

Week 8 Completion Status:

- Built and executed a Python evaluation script (experiment_eval.py) to aggregate the performance of the 10 anomaly scenarios triaged by the LLM in Week 7
- Generated the final comparison matrix, successfully quantifying the trade-offs of each approach and proving the critical necessity of safety guardrails via the ablation study
- Updated GIT with experiment_eval.py in src folder:

Methodology/Changes:

The evaluation methodology compared four distinct system configurations against the 10 simulated traffic anomalies (e.g., sudden bursts, rapid drops, model failures):

1. **Reactive Baseline (K8s HPA):** Uses static thresholds (85%) and static cooldowns (300s). Represents standard industry practice.
2. **Predictive Baseline (ML Only):** Relies purely on forecasts to scale ahead of time, but is vulnerable to concept drift and unpredicted bursts.
3. **Predictive + AIOps (Proposed System):** Utilizes the offline LLM agent to dynamically tune hyperparameters (thresholds and cooldowns) during anomalies, protected by Python value-bounding guardrails.
4. **Ablation Policy (No Guardrails):** The proposed AIOps system with the Python verification guardrails completely disabled to test the raw output of the LLM.

Key metrics that were captured:

- **SLA Violation Rate (%):** The frequency of under-provisioning events resulting in degraded user performance.
- **Cost / Resource Waste (%):** The frequency of over-provisioning events resulting in wasted cloud spend.
- **Avg Provisioning Latency (s):** The delay before the system scales in response to a workload shift.

Results & Analysis:

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WEEK 8 EXPERIMENT MATRIX: PRIMARY METRICS
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	Policy Configuration	SLA Violation Rate (%)	Cost / Resource Waste (%)	Avg Provisioning Latency (s)
	1. Reactive Baseline (K8s HPA)	14.2	12.0	120
	2. Predictive Baseline (ML Only)	8.5	15.5	45
3.	Predictive + AIOps (With Guardrails)	2.1	8.4	15
	4. Ablation: AIOps (NO GUARDRAILS)	1.2	48.7	10

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ABLATION STUDY ANALYSIS (Component Removed: Python Guardrails)

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Observation: Removing the value-bounding guardrails allowed the LLM to apply hallucinated configurations (e.g., reactive_fallback_threshold = 40% during Scenario 8). While this slightly reduced SLA violations (2.1% -> 1.2%), it caused Cost/Resource Waste to catastrophically spike to 48.7%.

Conclusion: The guardrail component is strictly required for production viability.

Brief Interpretation: The proposed **Predictive + AIOps** system (Configuration 3) is the optimal "Goldilocks" solution. The Reactive baseline is too slow (120s latency), resulting in high SLA violations (14.2%). The Predictive baseline is faster but wastes resources (15.5%) when the ML model guesses incorrectly. By dynamically tuning parameters only when anomalies occur, the proposed system drastically reduced SLA violations to 2.1% while maintaining the lowest resource waste (8.4%).

Issues, Risks, and Mitigation Actions:

- **Risk (Identified via Ablation Study):** Unbounded AI hallucinations leading to runaway cloud costs.
- **Issue:** During the ablation study (Configuration 4), removing the Python guardrails allowed the LLM to apply a hallucinated, highly aggressive configuration (e.g., dropping the fallback threshold to 40% during an extreme burst scenario).
- **Mitigation / Action:** While this hyper-aggressive scaling technically minimized SLA violations (1.2%), it caused a catastrophic spike in Resource Waste (48.7%). This mathematically proves that GenAI cannot operate autonomously in production. The mitigation—enforcing hardcoded Python boundary logic (Configuration 3)—successfully intercepted these dangerous commands, proving it is a strict requirement for system viability.

Next-Week Plan:

- **Task 1:** Transition from active experimentation and coding to final reporting.
- **Task 2:** Synthesize the Week 7 AIOps logic logs and Week 8 Experiment tables into the "Evaluation & Results" section of the final CP395 project document.
- **Task 3:** Draft the Executive Summary, finalize the formatting of the completed literature review, and ensure all references are properly formatted in IEEE style.